

THE COMPLETE END-TO-END PHYSICAL AGENT LIFECYCLE

From Natural Language Instruction to 3D Affordances, Expiring Leases, Action Chunks, Safety Barrier Firing & Physical Consequence

UNTRUSTED COGNITIVE PROPOSAL REALM · LINUX MPU / NPU (0.5–50 Hz)

1 · PROMPT & SENSE

Goal Ingestion

- Open-vocabulary text
- Photon DMA ring buffers
- Microsecond PTP stamps

Input: (text, I_0)
QoS: PTP t_{mid}

2 · 3D PERCEPTION

Affordance Grounding

- DINOv2 / ViT Patch tokens
- 3D Bounding Hulls B_{3D}
- Surface Normals n_{hat}

Tokens: $o_t \in R^{(N \times D)}$
Frame: SE(3) Targets

3 · WORLD MODEL

SE(3) Frame Tree

- Latent JEPA transition
- Transform Buffer query
- State TTL Expiration

State: z_t, Σ_t
Tracking: $d_{drift} \leq \delta$

4 · INTENT LEASE

VLM Semantic Lease

- Kinematic Reachability
- Time-to-Live TTL = 800ms
- Drift abort boundary

Lease: L_{intent}
Auth: 1 Hz Untrusted

5 · ACTION CHUNKS

Trajectory Synthesis

- Diffusion Policy / ACT
- $H=16$ waypoints
- C^2 quintic spline blend

Chunk: $A(t:t+H)$
Rate: 20–50 Hz

SHARED SRAM MAILBOX · PROPOSAL–PERMISSION SPLIT

6 · 1000 Hz BARE-METAL MCU REFLEX

Safety Barrier Quadratic Program Projection

- Control Barrier Function (CBF): Solves $\min \|u - p_{tj}\|^2$ s.t. $\dot{h} \geq -\alpha(h)$ in $< 65 \mu s$
- Dynamic stopping clearance audit: $d_{stop}(v) \leq d_{clearance}$
- Hardware Watchdog & IEC 60204-1 emergency fallbacks (Cat 0/1/2 safe stops)

PWM Output: $u_{safe}(t)$ latched to gate drivers in $< 180 \mu s$ (Slack = 820 μs)

7 · PHYSICAL CONSEQUENCE & FEEDBACK

State Mutation $W_t \rightarrow W_{t+1}$ & Feedback O_{t+1}

- Phase currents energize motor windings, accelerating mass m with momentum $p = mv$.
- Irreversible physical consequences: you cannot `ctrl+z` kinetic energy.
- Physical action mutates the environment, producing endogenous new photons O_{t+1} .

Closed Loop: $A_t \rightarrow W_{t+1} \rightarrow O_{t+1}$ feeds back to Step 1 Sensing